Question Paper

Back

MANIPAL ACADEMY OF HIGHER EDUCATION

III SEM. BTECH. (CSE/IT/CCE/CSE-AIML/DSE)
SCHOOL OF COMPUTER ENGINEERING
MID TERM EXAMINATION, SEPT. 2025
SUBJECT: DATA ANALYTICS [CSS 2103]

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Marks: 30 **Duration: 90 mins.**

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Answer all the questions.

1) The heights of six randomly chosen sailors are in inches: 63, 65, 58, 69, 71, and 72. The heights of ten randomly chosen soldiers are in inches: 61, 62, 65, 66, 69, 69, 70, 71, 72, and 73. Do these figures indicate that the soldiers are on average, shorter than the sailor? Statistically test at 5% significance level (4)

2) Given the dataset: [5, 7, 8, 5, 6, 7, 120, 6, 8, 7]. Apply the Interquartile Range (IQR) method to identify any outliers. Show the steps involved in calculating Q1, Q3, IQR, and the outlier boundaries, to identify the outliers. (3)

3) Given the dataset:[5, 6, 7, 8, 5, 6, 7, 8, 50, 60]. Apply the skewness formula

$$Skewness = \left(\frac{\sqrt{n(n-1)}}{n-2}\right) \times \frac{\frac{1}{n}\sum_{i=1}^{n} (x_i - \bar{x})^3}{\left(\frac{1}{n}\sum_{i=1}^{n} (x_i - \bar{x})^2\right)^{3/2}}$$ (3)

to calculate the skewness score of the dataset. Show all the steps and interpret whether the data is symmetrical, positively skewed, or negatively skewed.

4) Explain the methods to transform the data so that it closely approximates a normal distribution. (3)

5) Explain the different methods of filling the missing values for the numerical data. (3)

6) The table shown has the data collected during the outbreak of smallpox.

	Attacked	Not Attacked
Vaccinated	31	469
Not Vaccinated	185	1315

(3)

Test the effectiveness of vaccination in preventing smallpox at 5% significance level

7) (3)

	Using the dataset of student heights (in cm) = {160, 165, 170, 175, 180} and weights (in kg) = {55, 58, 62, 65, 70}, calculate the covariance and interpret whether the relationship indicates a positive, negative, or no relation between the variables.	
8)	A faculty member wants to analyse the performance of the students in the course he offers. The data set taken for analysis contains the following variables: Sl. No., Registration Number, Test1 marks, Test2 marks, Test3 marks, Final marks, and CGPA. Identify and explain the most suitable Dimensionality Reduction technique when the interpretability of the transformed result is not a concern. Further, identify the irrelevant features in this dataset.	(2)
9)	A researcher collects the following information from 20 individuals: Gender, Age, Height (in cm), Weight (in kg), Blood Group, Mobile Number, Education Level (High School, Undergraduate, Postgraduate, PhD), Monthly Income. Classify each variable into: (i) Discrete or Continuous (ii) Nominal, Ordinal, Interval, or Ratio.	(2)
10)	A university is preparing student data for performance prediction. You observe the following problems: 1. The attribute attendance_percentage has missing values for several students. 2. The attribute grades are represented using both letter grades (A, B, C) and numerical scores (85, 72, 60). Explain the preprocessing steps you would apply to handle these issues.	(2)
11)	Explain the Entity Identification problem. In a student dataset, both Roll Number and Registration Number are used as identifiers. Explain which attribute should be selected as the primary entity identifier during preprocessing and why.	(2)

^ Top